BUI SON ANH

AI / ML Engineer · LLM · RAG · Multi-Agent Systems
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SUMMARY

AI Engineer specializing in production LLM systems — RAG pipelines, agentic workflows, and LLM evaluation/observability. Currently the lead AI engineer on a customer-support RAG chatbot shipped to production at PIXTA, with hands-on experience building multi-agent systems and a research background in computer vision and deep learning. Strong focus on shipping reliable, measurable AI products end to end, from prototyping through deployment and monitoring.

WORK EXPERIENCE

AI Engineer — Full-time *Jun 2025 – Present*

PIXTA Vietnam Co., Ltd. · AI Research

- Lead AI engineer on the **CS Chatbot (RAG → agent)** — designed and built a production customer-support assistant (Milvus vector search + LLM generation), owning nearly the entire build and shipping the MVP to production, then iterating through v0.4 → v0.7.
- Improved answer quality from ~29% coverage / 38.5% fallback at launch to ~51% bot coverage with 88% accept rate, via iterative prompt and knowledge-base work, multi-query retrieval with Reciprocal Rank Fusion (RRF), chunk-size tuning (600 → 300), and migrating the generation model (Gemini → Claude Sonnet).
- Cut average production latency from ~12s to ~5–6s by re-architecting the RAG pipeline into an agent that retrieves only when needed, adding rule-based retrieval for deterministic cases, constraining the agent to one tool call per turn via middleware, fixing streaming/ghost-latency issues, and setting a hard timeout.
- Redesigned the document-ingestion pipeline — converting hard-to-parse FAQ tables into clean markdown / structured bullets and surfacing source URLs alongside text — significantly improving retrieval accuracy across categories.
- Built an end-to-end LLM evaluation and observability stack (Langfuse + LLM-as-judge: correctness, faithfulness, retrieval metrics) used as an automated release gate with regression testing across categories and 100% coverage on critical rules.
- Defined the project's shared success metric — *Bot Ability = bot answers / (bot answers + form contacts)* — on a Redash dashboard, and designed a methodology to quantify human-resource savings that isolates the chatbot's effect from natural contact decline using multi-year historical data.
- Hardened the bot against real production attacks, building a filter layer to block SQL-injection and spam attempts in ~0.5 days with no SRE involvement; improved a Japanese-language support prompt (buyer/creator user-type disambiguation, fixing tool-use hallucination).
- Took ownership of the full product lifecycle, self-managing dev/deploy and release to production (Docker, Vertex AI) — and was delegated production-release rights — to cut release time and reduce dependency on SRE.
- Built a **semantic photographer-matching PoC for fotowa** — proposed the approach and delivered a working demo within a < 4-week timeline, combining portfolio-style and personality signals with a human-evaluation methodology for matching accuracy.
- Delivered a **Conversational Search PoC** in 2 days, including a feature that suggests search ideas to users; received positive feedback from stakeholders.
- Drove internal knowledge sharing — host of the team's seminar series, delivered a hands-on workshop on the Agno multi-agent framework (live chess multi-agent challenge with FastAPI/WebSocket visualization), and seminars on modern LLM tooling; Langfuse was adopted as the team's core evaluation layer for LLM projects.

AI Engineer — Intern *Oct 2024 – Jun 2025*

PIXTA Vietnam Co., Ltd.

- Researched and optimized Vision-Language Models (VLMs) and vector-based search for improved retrieval accuracy and performance.
- Ran benchmarking and indexing optimization; contributed to system documentation and enhancements.

TECHNICAL SKILLS

LLM / Agents: RAG, LLM evaluation (LLM-as-judge), Agno, LangGraph, LangChain, AutoGen, MCP, mem0, prompt engineering, Claude API, Gemini

Retrieval / Data: Milvus, FAISS, vector search, learning-to-rank, Redis, SQL, BigQuery

Vision / ML: PyTorch, Hugging Face Transformers, VLMs (BLIP-2, LLaVA, Qwen-VL, MiniCPM-V), CLIP, YOLO, PaddleOCR

MLOps / Infra: Docker, FastAPI, Vertex AI, AWS, Langfuse, CI/CD, production monitoring
Languages: Python; English (IELTS 7.5); Vietnamese (native)

EDUCATION

B.Sc. in Computing — First Class Honours *Aug 2022 – Mar 2026*
FPT Greenwich University
Additional courses: AIO2024 (AI Viet Nam), Machine Learning Specialization (Coursera), Grounding LLMs for Reliable Apps (GDG 2025)

SELECTED RESEARCH & PUBLICATIONS

LJPA: Landmark-Guided Jigsaw Puzzle Autoencoders for Facial Expression Recognition · *CSoNet 2025 (accepted)*

- First author. Self-supervised method using shuffled facial patches for expression-aware representations; +5.4–6.0% accuracy over ResNet18 on AffectNet, RAF-DB, and FER-Plus.

An Enhanced Deep Learning Model for Image Classification and FER Using Self-Distillation · *ACOMPA 2024*

- Co-author. ResNet18-Lite (self-distillation + CBAM) for resource-constrained devices; SOTA on FER2013, FERPlus, CIFAR100. Highest prize, IT division, FPT Research Festival 2024. [paper]

Additional IEEE Access publications (Research Assistant)

- Vision-based industrial anomaly detection; depth-assisted monocular object pose estimation; multimodal hand pose estimation. [1] [2]

ACTIVITIES & SERVICE

- Judge, SVNCKH 2025–2026 Student Scientific Research Conference (HUST) — scored and reviewed student papers across novelty, methodology, clarity, results, and impact.
- Host of the AI Research team's internal seminar series — organize sessions and a multi-criteria scoring framework for participant submissions.

AWARDS

- Champion, IT division — FPT Research Festival 2024.
- Semi-finalist — FPT Edu Hackathon 2024.
- 30% Scholarship & Top 3 Students, Spring 2023 — Greenwich Vietnam.
- Second place, Junior Group — Greenwich Coding Challenge 2023.